

Lab 5: Static IP Configuration on a Linux Machine using Graphical User Interface (GUI)

Lab Date: 1 July 2026 | Deadline: 6 July 2026

BSc.CSIT 4th Semester | New Summit College | Computer Network

Objective: To understand the basic networking parameters of a machine and configure a static IP address on a Linux system using the Network Settings GUI.

1. Theory

1.1 IP Address

An IP (Internet Protocol) address is a unique numerical label assigned to every device connected to a network. It allows a device to be identified and located so that data can be routed to and from it correctly. An IPv4 address is written as four decimal numbers separated by dots (e.g. 192.168.1.10), each ranging from 0–255.

Static IP Address: An IP address that is manually configured on a device and does not change unless it is manually reconfigured. It is commonly used for servers, printers and network devices that need to be reachable at a fixed address.

Dynamic IP Address: An IP address that is automatically assigned to a device by a DHCP (Dynamic Host Configuration Protocol) server, and which may change each time the device reconnects to the network. It requires little manual configuration.

1.2 Subnet Mask

A subnet mask (e.g. 255.255.255.0) is used along with an IP address to divide it into a network portion and a host portion. It tells the device which part of the IP address identifies the network and which part identifies the specific host on that network.

1.3 Default Gateway

The default gateway is the IP address of the router that a device sends traffic to when the destination is outside its own local network. It acts as the exit/entry point between the local network and other networks such as the Internet.

1.4 DNS Server

A DNS (Domain Name System) server translates human-readable domain names (e.g. www.google.com) into the numeric IP addresses that computers use to identify each other, allowing users to access websites without memorizing IP addresses.

2. Implementation Sequence

- a **Test the device's connectivity and current IP using the terminal** — Open a terminal and run `ip a` (or `ifconfig`) to view the current IP address, and `ping google.com` to confirm the device is connected to the network.
- b **Accessing Network Settings** — Open *Settings* → *Network* (or the Network Manager applet) and select the active wired/wireless connection, then open its settings/gear icon.
- c **Change the IP statically using the GUI** — Under the *IPv4* tab, switch the method from "Automatic (DHCP)" to "Manual", then enter the desired Address, Netmask, Gateway, and DNS server values, and click Apply/Save.

- d **Re-test connectivity and IP using the terminal** — Reconnect the network interface (toggle it off/on or run `nmcli connection up <name>`) and run `ip a` and `ping` again to confirm the new static IP has taken effect and the connection is still working.

■ *[Insert Screenshot Here] — terminal output of ip a before configuration*

■ *[Insert Screenshot Here] — Network Settings window / adapter list*

■ *[Insert Screenshot Here] — IPv4 manual configuration screen with entered values*

■ *[Insert Screenshot Here] — terminal output of ip a and successful ping after configuration*

Conclusion

This lab helped in understanding how IP addressing, subnet masks, default gateways and DNS servers work together to enable network communication. By configuring a static IP address through the Linux GUI, the difference between a dynamically assigned and a manually assigned IP address, and the practical steps involved in switching between the two, was clearly understood and verified using terminal commands.

Lab 6: Static IP Configuration on a Linux Machine using Command Prompt / Terminal (CLI)

Lab Date: 1 July 2026 | Deadline: 6 July 2026

BSc.CSIT 4th Semester | New Summit College | Computer Network

Objective: To understand the basic networking parameters of a machine and configure a static IP address on a Linux system using the Command Line Interface (CLI).

1. Theory

1.1 Command Line Interface (CLI)

The Command Line Interface is a text-based interface that allows a user to interact with the operating system by typing commands instead of using graphical menus. On Linux, this is the terminal (shell), which offers precise and scriptable control over system and network configuration.

1.2 ip / ifconfig Command

The `ip` command (the modern replacement for the older `ifconfig`) is used to display and configure network interfaces on Linux — for example, `ip a` shows all IP addresses assigned to the machine's interfaces, and `ip link` shows the status of each network interface.

1.3 nmcli / netplan Command

`nmcli` is the command-line tool for Network Manager, used to view and modify network connections (equivalent to Windows' `netsh`). On Ubuntu Server systems, `netplan` configuration files (YAML, under `/etc/netplan/`) can also be edited and applied with `sudo netplan apply` to set a static IP.

2. Implementation Sequence

- Test the device's connectivity and IP using CLI** — Run `ip a` to view current IP settings and `ping google.com` to confirm the current network connection is working.
- Select the active adapter** — Run `nmcli device status` (or `ip link`) to list all interfaces and identify which one is currently active/connected (state: connected).
- Change the IP statically using CLI** — Configure the static values using `sudo nmcli con mod <connection-name> ipv4.addresses <IP/prefix> ipv4.gateway <gateway> ipv4.dns <dns> ipv4.method manual`, then bring the connection back up with `sudo nmcli con up <connection-name>`.
- Re-test connectivity and IP using CLI** — Run `ip a` again to confirm the new static address has been applied, and `ping` a known host to confirm connectivity is still working.

■ *[Insert Screenshot Here] — terminal output of ip a and ping before configuration*

■ *[Insert Screenshot Here] — output of nmcli device status showing the active adapter*

■ *[Insert Screenshot Here] — nmcli con mod command used to set the static IP*

■ *[Insert Screenshot Here] — terminal output of ip a and ping after configuration*

Conclusion

This lab provided hands-on experience in configuring a static IP address purely through the terminal, reinforcing the concepts of IP addressing, subnetting, gateways and DNS from a command-line

perspective. It also highlighted the advantage of the CLI over the GUI in terms of speed and scriptability when managing network configuration on Linux systems.